Course: Celestial Mechanics   Instructor: David M. Hernandez   TA: Vincent Gene L. Otero

Restricted Three - Body Problem

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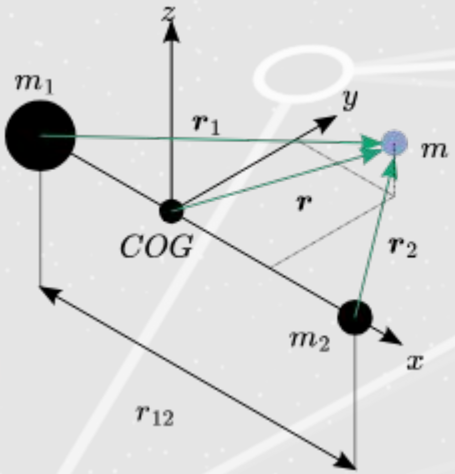
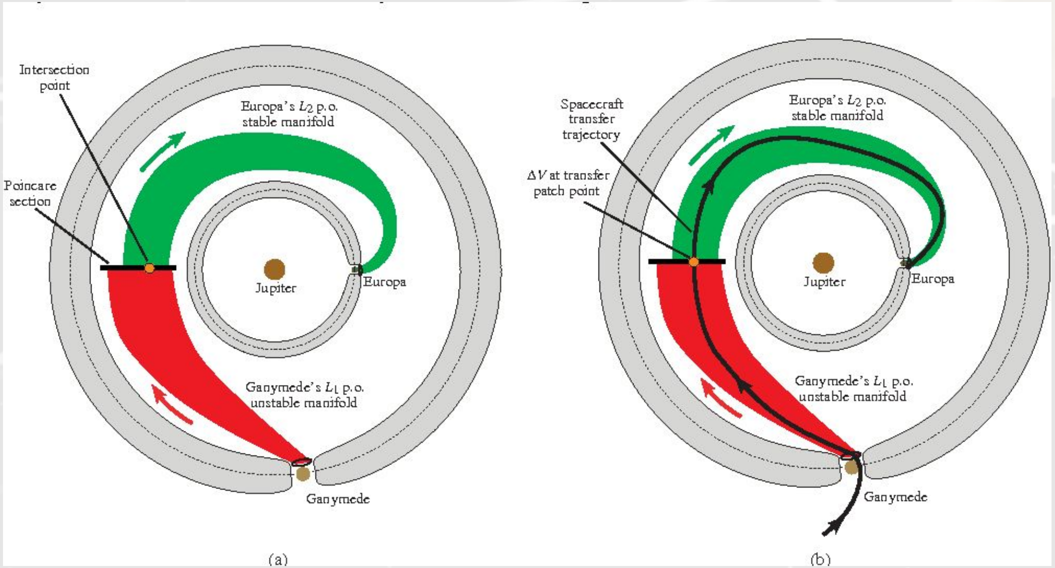
Objectives

- Identify and classify as many distinct classes of orbits as possible by integrating equations of motion.
- Test the stability criterion for triangular Lagrange points numerically.
- Vary initial conditions, the Jacobi constant, and the mass ratio to map the structure of the orbit.

Restricted Three-Body Problem

Two of the bodies move in circular, coplanar orbits around their common center of mass





G. Gómez et al., 2001

Framework Equations

Mass ratio: $$\mu = \frac{m_1}{m_0 + m_1}$$

Equations of motion in the synodic frame

$$\ddot{x} - 2n\dot{y} - n^2 x = -\left[\mu_1 \frac{x + \mu_2}{r_1^3} + \mu_2 \frac{x - \mu_1}{r_2^3}\right], \qquad (3.16)$$

$$\ddot{y} + 2n\dot{x} - n^2 y = -\left[\frac{\mu_1}{r_1^3} + \frac{\mu_2}{r_2^3}\right] y, \qquad (3.17)$$

$$\ddot{z} = -\left[\frac{\mu_1}{r_1^3} + \frac{\mu_2}{r_2^3}\right] z. \qquad (3.18)$$

Distances from the test object to primary mass

$$r_1^2 = (x + \mu_2)^2 + y^2 + z^2, \qquad (3.8)$$

$$r_2^2 = (x - \mu_1)^2 + y^2 + z^2, \qquad (3.9)$$

Lagrange Point Stability
~ 0.0385

$$\mu < \frac{1 - \sqrt{23/27}}{2}$$

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L Points

According to Murray & Dermott:

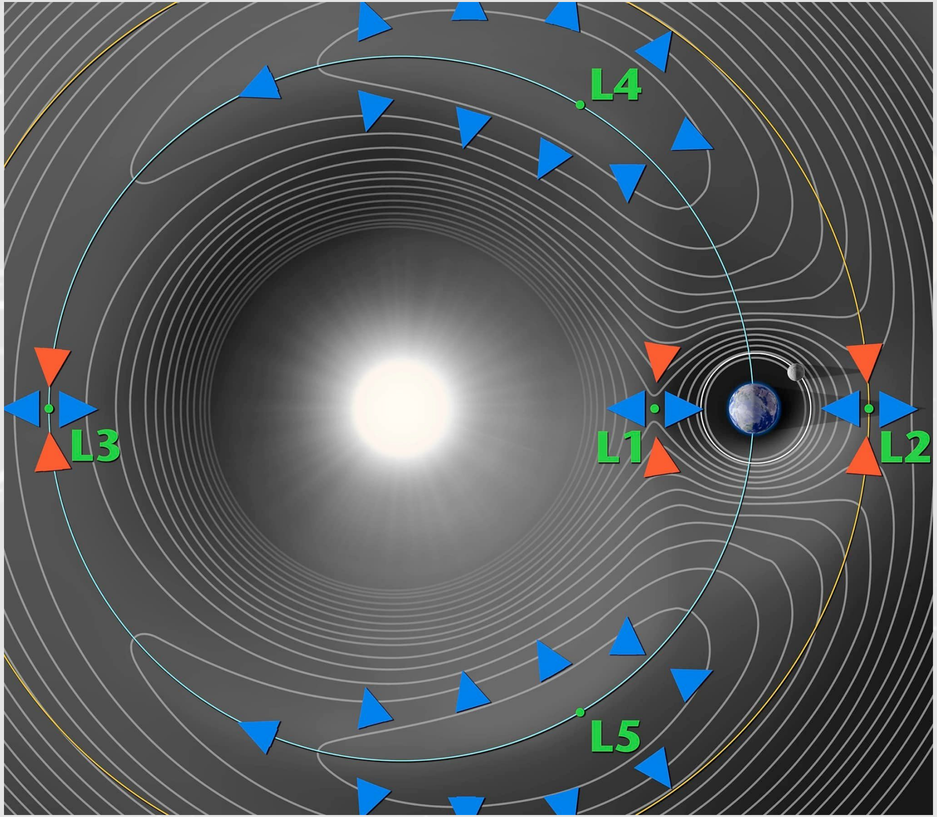
L1: $r_1 + r_2 = 1.$

L2: $r_1 - r_2 = 1$

L3: $r_2 - r_1 = 1$

L4: at the apex of an equilateral triangle with a base formed by the line joining the two masses

L5: another equilibrium point located below the same line, also lying at the apex of an equilateral triangle



Leapfrog

$\boxed{x_n, v_n}$

↓

$\boxed{a_x, a_y, a_z}$

↓

$$\boxed{v_{x\frac{1}{2}} = v_x + a_x \cdot \frac{\Delta t}{2}}$$

↓

$$\boxed{x_{new} = x + v_{x\frac{1}{2}} \cdot dt}$$

↓

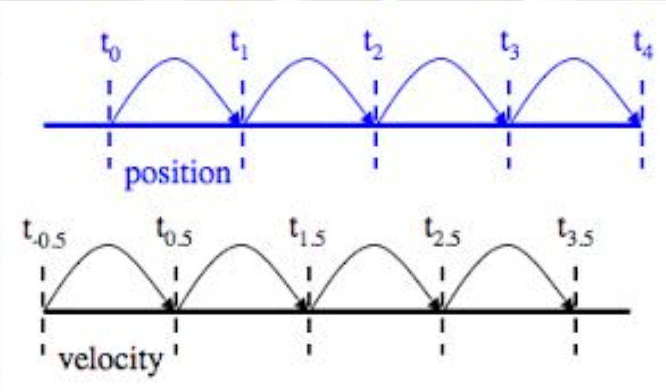
$\boxed{\text{new } a_x, a_y, a_z}$

↓

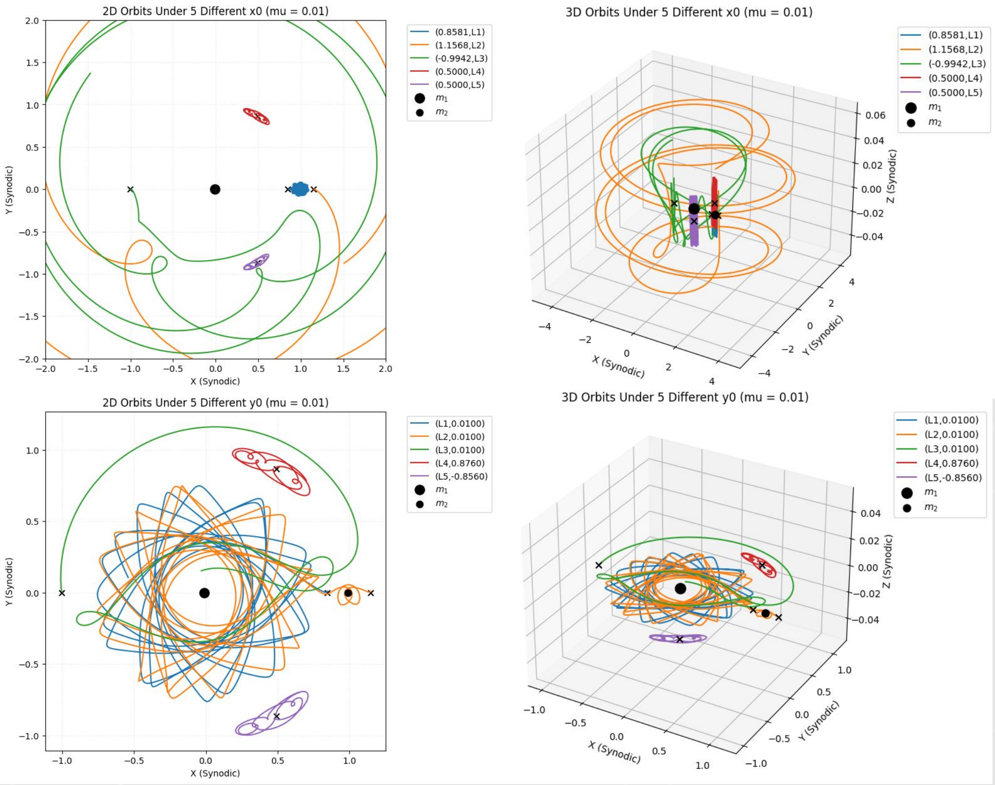
$$\boxed{v_{x\,new} = v_{x\frac{1}{2}} + a_{x\,new} \cdot \frac{dt}{2}}$$

↓

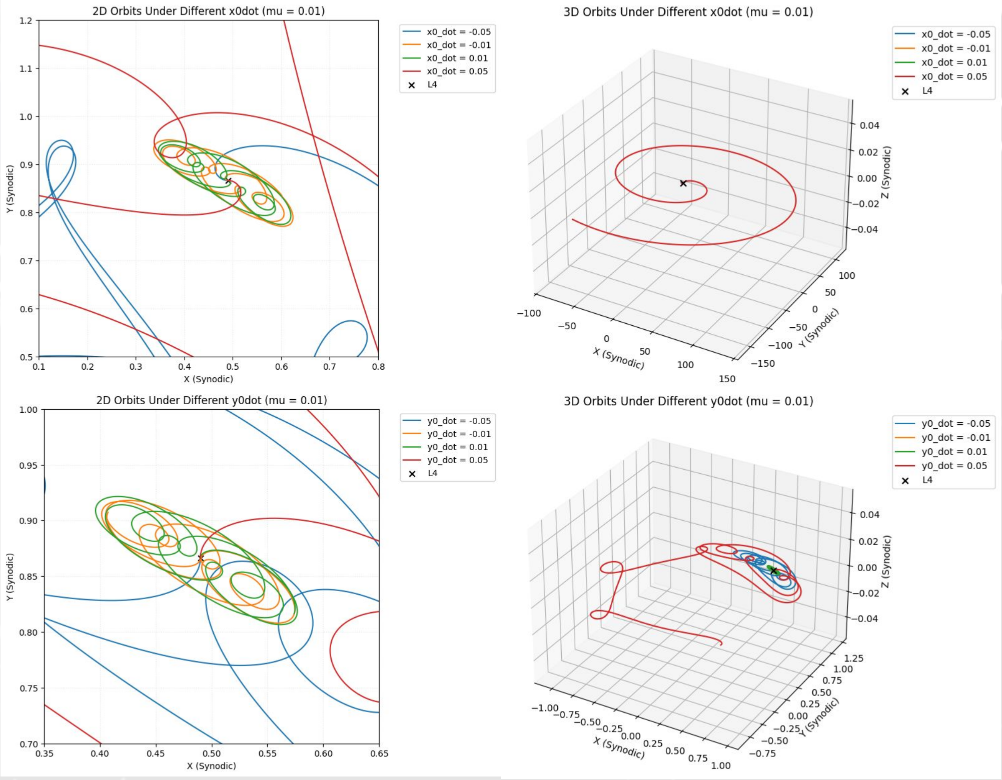
$\boxed{x_{new}, v_{new}}$



Varying Initial Position



Varying Initial Velocity



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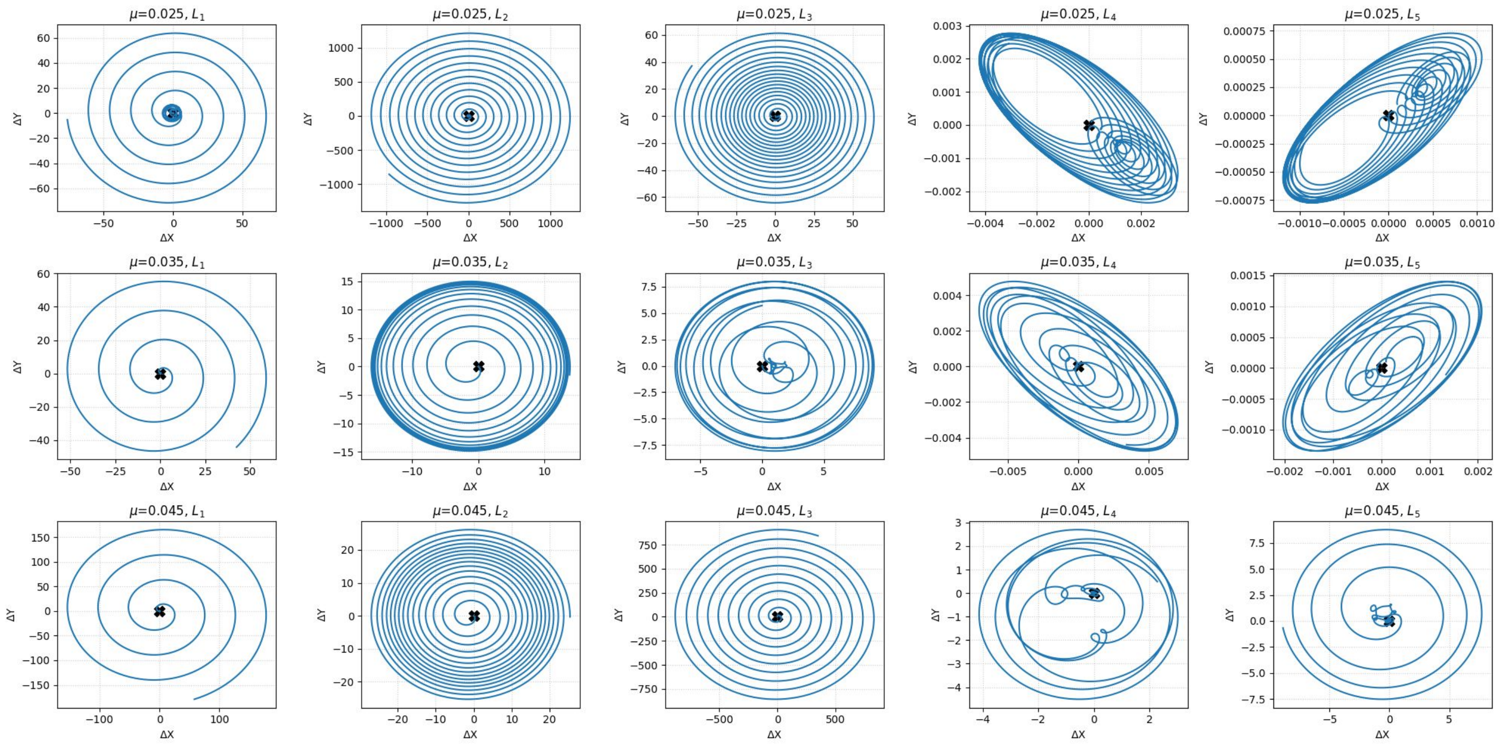
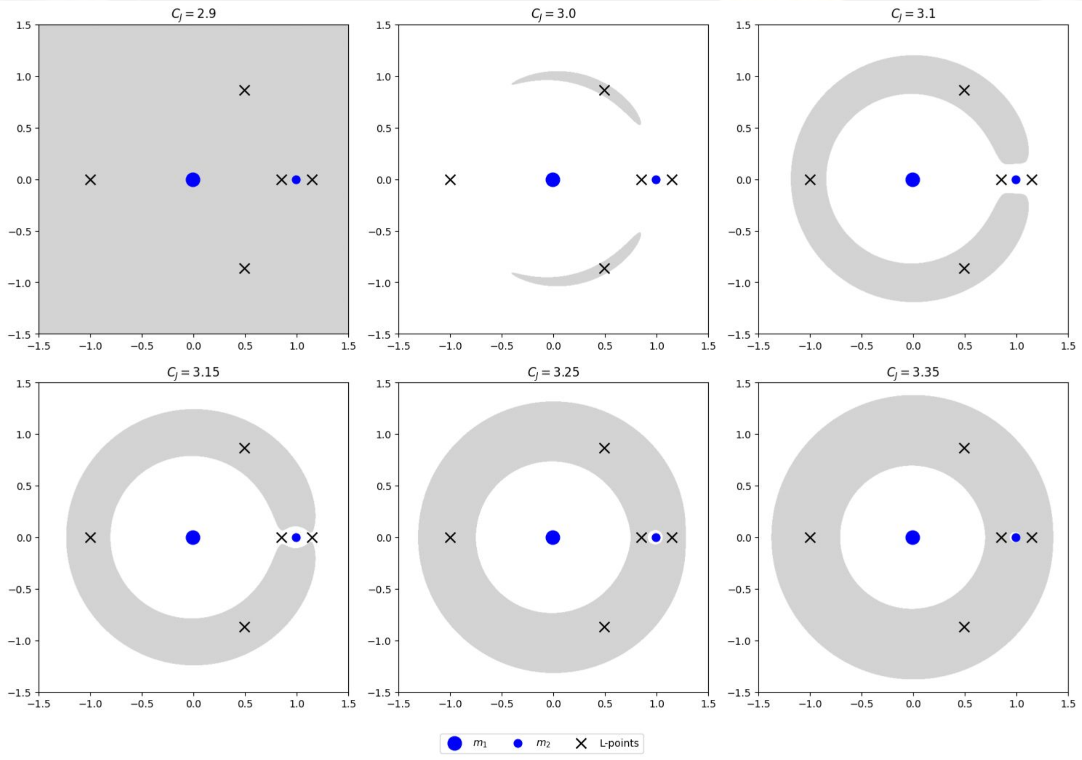
Stability

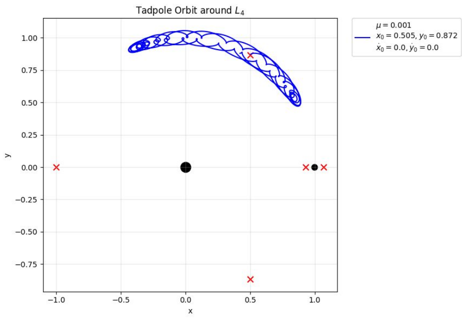


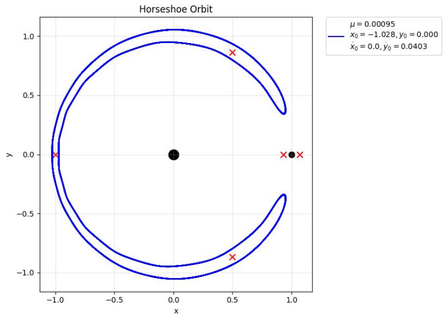
μ=0.025, L_1
μ=0.025, L_2
μ=0.025, L_3
μ=0.025, L_4
μ=0.025, L_5
μ=0.035, L_1
μ=0.035, L_2
μ=0.035, L_3
μ=0.035, L_4
μ=0.035, L_5
μ=0.045, L_1
μ=0.045, L_2
μ=0.045, L_3
μ=0.045, L_4
μ=0.045, L_5
ΔX
ΔY

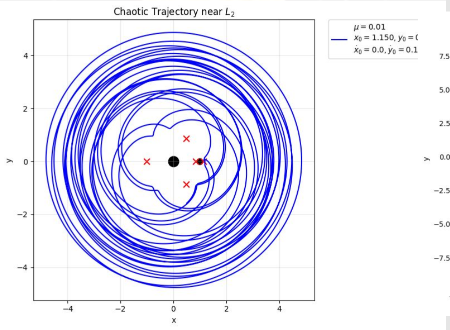
Jacobi Constant and Zero-Velocity Regions

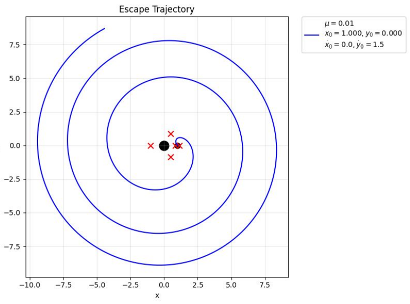


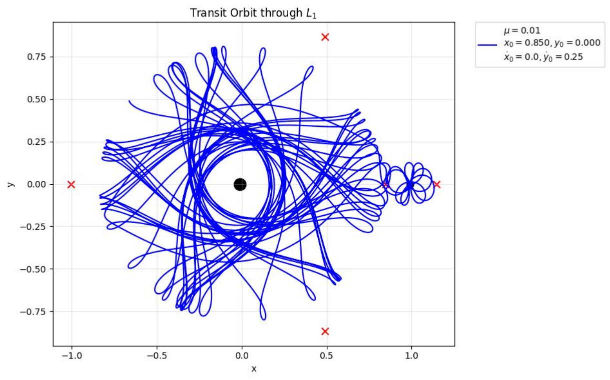
Classification of Orbits

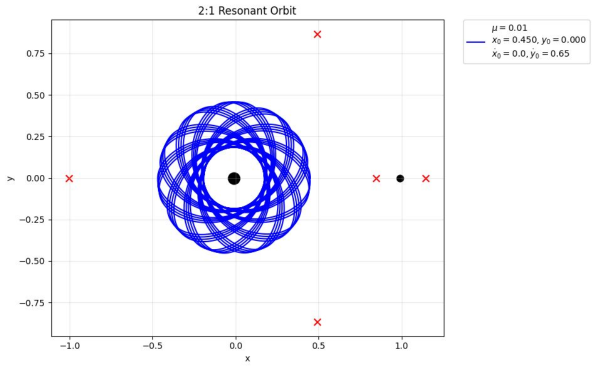


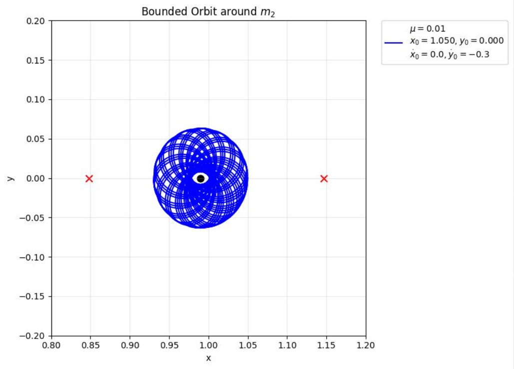












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Interactive Simulator

https://codepen.io/La-Th-La/pen/PwbeJap

Conclusion

- The Jacobi Constant C_J dictates zero-velocity curves, defining whether a travelling object/particle is trapped or escaping.
- The critical mass ratio limit mu < 0.03852.
- L1, L2, L3 are energetic gateways that accelerate particles into chaotic or escape paths.
- L4 and L5 are gravitational traps that keep particles bounded.

THANK YOU!